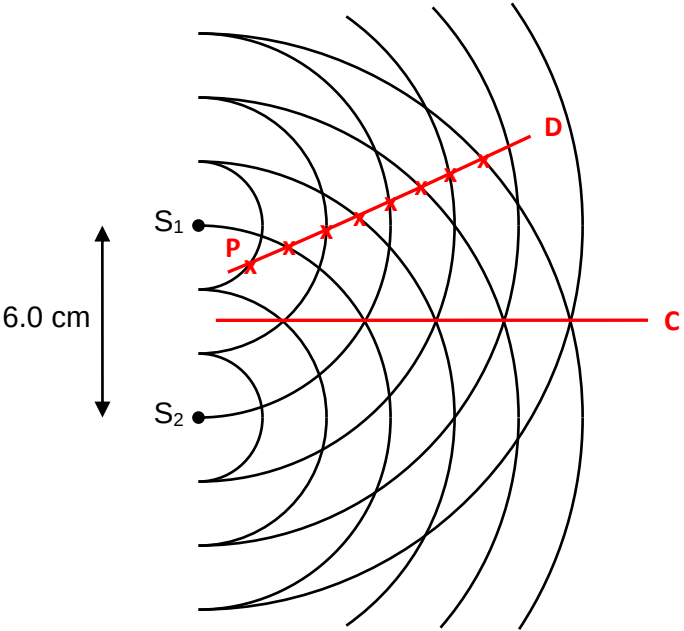


7	(a)	(i)	progressive means there is energy transfer in the direction of travel of wave (accept the lecture notes version)	B1
		(ii)	same or similar amplitude either unpolarised or polarised in the same plane	B1 B1
 Fig. 7.1				
		(iii)	line through 3 intersections where path difference is 0 and label C	B1
		(iv)	point where path difference = 1.5λ and label P (any of the above) line through P and 3 other points and label D	B1 B1
		(v)	$3\lambda = 6.0 \text{ cm}$ B1 $\lambda = 2.0 \text{ cm}$	
	(b)	(i)	path difference $S_2P - S_1P = \sqrt{8.4^2 + 15^2} - \sqrt{2.4^2 + 15^2}$ M1 $= 2.0 \text{ cm} = \text{one wavelength}$ A1	
		(ii)	path difference $= 1.5\lambda = 3.0 \text{ cm}$	B1
		(iii)	$\sqrt{(OY + 3)^2 + 15^2} - \sqrt{(OY - 3)^2 + 15^2} = 3.0$ M1 $OY = 8.8 \text{ cm}$ A1	

		(iv)	intensity at O due to 1 source $= \frac{P}{4\pi r^2} = \frac{20}{4\pi(0.03^2 + 0.15^2)}$ $= 68 \text{ W m}^{-2}$ M1 A0	
		(v)	$I \propto A^2$ at maxima, amplitude doubled so intensity = $4 \times 68 = 272 \text{ W m}^{-2}$	B1
	(c)	(i)	(antinode to antinode distance = $\frac{\lambda}{2} = \frac{2.0}{2} = 1.0 \text{ cm}$)	B1
		(ii)	rate of fluctuations = $\frac{\text{speed of detector}}{\text{internodal distance}} = \frac{3.0 \text{ cm s}^{-1}}{1.0 \text{ cm}} = 3.0 \text{ s}^{-1}$	B1
	(d)	(i)	limiting angle of resolution = $\frac{\lambda}{b}$ $= \frac{500 \times 10^{-9}}{0.10}$ $= 5.0 \times 10^{-6} \text{ rad}$ C1 A1	
		(ii)	angle of separation of sources $\approx \frac{6.0 \times 10^{-2}}{15 \times 10^3} = 4.0 \times 10^{-6} \text{ rad}$	B1
		(iii)	angle of separation of sources < limiting angle of resolution, so cannot resolve	B1

8	(a)	Since the spring constant k and the mass m are both constant	
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